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Exploratory Data Analysis of Large-Scale Retail Transaction Data Using Univariate, Bivariate, and Multivariate Statistical Techniques

Introduction

Exploratory Data Analysis (EDA) is a statistical approach used to summarize, visualize, and understand datasets before applying predictive models.

This project performs detailed EDA on a large retail transaction dataset containing 100,000 records and 18 variables. The focus is on statistical distribution analysis, relationships between variables, and identifying unusual patterns.

Situation Overview:

The primary objective of this project is to:

- Perform **univariate, bivariate, and multivariate analysis**
- Detect and interpret **outliers**
- Measure **correlation between numerical variables**
- Analyze **skewness and kurtosis**
- Extract statistically meaningful business insights

Dataset Overview

Rows: 100,000

Columns: 18

Data Types: Categorical, Numerical, Date

Download the dataset [“retail_large_dadataset”](#) and upload the dataset in your project environment.

Steps in EDA for Real Estate Pricing:

Univariate Analysis

Univariate analysis examines one variable at a time.

4.1 Numerical Variables

For each numerical column, the following were computed:

- Mean
- Median
- Standard Deviation

- Minimum & Maximum
- Skewness
- Kurtosis

Example Insights:

- **product_price** showed positive skewness → more lower-priced products than expensive ones.
- **final_price** exhibited high kurtosis → presence of extreme values (bulk purchases).
- **discount_percentage** was approximately uniformly distributed between 0–30%.

Categorical Variables

Frequency distributions were analyzed for:

- product_category
- customer_segment
- payment_method
- return_status

Example findings:

- Majority of customers belonged to the "Regular" segment.

- Electronics category had the highest transaction count.
 - Return rate was approximately 15%.
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Outlier Detection

Outliers were detected using:

- Boxplots
- Interquartile Range (IQR) method
- Z-score method

Observations:

- Extreme values were present in **product_price**.
- High outliers in **final_price** corresponded to:
 - High quantity purchases
 - Premium product categories
- Some long delivery times were identified as operational outliers.

Outliers were not removed immediately, as they may represent valid high-value transactions.

Bivariate Analysis

Bivariate analysis studies relationships between two variables.

Numerical vs Numerical

Correlation analysis was performed using Pearson correlation.

Key relationships:

- Strong positive correlation between:
 - product_price and final_price
 - quantity and final_price
- Weak correlation between:
 - age and spending
- Slight positive correlation between:
 - delivery_days and return_status (encoded)

Numerical vs Categorical

Examples:

- Customer segment vs average final_price

- Product category vs return rate
- Shipping type vs delivery_days

Findings:

- Premium customers had higher average spending.
 - Electronics had slightly higher return rates.
 - Express shipping reduced delivery days significantly.
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Correlation Matrix Analysis

A correlation heatmap was generated for numerical variables.

Observations:

- final_price strongly correlated with:
 - product_price
 - quantity
 - discount_percentage (negative correlation)
- delivery_days had low correlation with spending

- discount_percentage negatively correlated with revenue per unit

No multicollinearity issues were detected among independent features.

Multivariate Analysis

Multivariate analysis examines interactions between three or more variables.

Examples:

1. Product Category $\times$ Discount $\times$ Final Price
→ Electronics with moderate discounts generated highest revenue.
2. Customer Segment $\times$ Age $\times$ Spending
→ Premium customers in age group 30–45 spent the most.
3. Delivery Days $\times$ Shipping Type $\times$ Return Status
→ Standard shipping with longer delivery times increased return probability.

These patterns highlight operational and pricing strategy insights.

Skewness and Kurtosis Interpretation

Skewness

- Positive skew observed in:

- product_price
- final_price
- Indicates majority of transactions are small-to-medium sized with few large transactions.

Kurtosis

- High kurtosis in:
 - final_price
- Suggests heavy tails (presence of extreme high-value purchases).

Understanding skewness and kurtosis helped identify:

- Distribution asymmetry
- Extreme purchase behaviors
- Potential business risk areas

Key Statistical Findings

- Data showed right-skewed revenue distribution.
- Revenue driven by limited high-value transactions.

- Moderate discounts maximize spending.
 - Outliers represent genuine premium purchases.
 - Return probability increases slightly with longer delivery times.
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Conclusion

This project successfully applied:

- Univariate statistical analysis
- Bivariate relationship analysis
- Multivariate interaction analysis
- Correlation study
- Outlier detection
- Skewness and kurtosis measurement

The analysis provides a strong foundation for predictive modeling and business decision-making.

In Python, Numpy is a package that includes multidimensional array objects as well as several derived objects.

Matplotlib is an amazing visualization library in Python for 2D plots of arrays.

Seaborn is on top of matplotlib for effective plot style

Pandas are used for data manipulation and analysis. So these are the core libraries that are used for the EDA process.

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